

Coffee Shop Sales Analysis

Project Overview

This project performs **Exploratory Data Analysis (EDA)** on a Coffee Shop Sales dataset using **Python, Pandas, Matplotlib**, and **Seaborn**. The objective is to analyze sales performance, customer purchasing behavior, product categories, and store-wise trends to generate meaningful business insights through statistical analysis and data visualization.

Objectives

- Analyze coffee shop sales data.
 - Identify high-performing store locations and product categories.
 - Study weekday-wise sales patterns.
 - Examine transaction quantity and pricing trends.
 - Generate business insights using visualizations.
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Technologies Used

- Python
 - Jupyter Notebook
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
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Dataset Features

The dataset contains information such as:

- Store Location
- Product Category
- Product Type

- Total Amount
 - Transaction Quantity
 - Unit Price
 - Weekday
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Analysis Performed

The following analyses were carried out:

- Generated descriptive statistics using `df.describe()`.
 - Performed correlation analysis with `df.corr()`.
 - Calculated total sales by store location using `groupby()`.
 - Created pivot tables to compare product categories and weekday sales across stores.
 - Conducted interactive analysis by selecting store location and product category.
 - Generated business insights using statistical summaries and aggregation techniques.
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Visualizations

Several visualizations were created to understand the data, including:

- Store Location Count Plot
- Product Category Count Plot
- Weekday Count Plot
- Store-wise Sales Bar Chart
- Average Transaction Quantity by Product Category
- Weekday Sales Line Plot
- Product Category Sales Distribution (Pie Chart)
- Store-wise Sales Distribution (Pie Chart)
- Histogram of Total Amount
- Point Plot

- Strip Plot ,Violin Plot
- Swarm Plot
- Scatter Plot
- Pair Plot
- Joint Plot
- KDE Plots

These visualizations help identify sales patterns, customer behavior, pricing trends, and product performance.

Key Insights

- Store performance varies significantly across locations.
 - Some product categories contribute more to total revenue than others.
 - Customer purchasing behaviour changes across weekdays.
 - Transaction quantity and unit price influence total sales.
 - Visualizations make it easier to identify trends and support business decision-making.
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Project Structure

Coffee-Shop-Sales-Analysis/

|— Dataset/

|— Notebook/

|— README.md

|— requirements.txt

Future Improvements

- Build a sales forecasting model.
- Create an interactive Power BI/Tableau dashboard.
- Perform customer segmentation.

- Develop a recommendation system for products.

Author

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